Adventist University of Central Africa
P.O. Box 2461 Kigali, Rwanda | www.auca.ac.rw | info@auca.ac.rw

BIG DATA ANALYTICS | FINAL PROJECT

E-Commerce Big Data Analytics Platform

with MongoDB, HBase and Apache Spark

Technical Report

Student: Uwizeye Christelle
Student ID: 101465
Course: Big Data Analytics
Submission: 11 November 2026

Project repository
https://github.com/uwizeyec/Big-Data-Analytics_Final-Project_Uwizeye-Christelle_101465

A reproducible analytics system that places e-commerce business entities in MongoDB, browsing sessions in HBase, and cross-store analytics in Apache Spark.

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1 Introduction and system architecture

This project implements an e-commerce analytics platform over a synthetic 90-day dataset containing 10,000 users, 5,000 products, 25 categories, 50,000 transactions, and 200,000 browsing sessions. It demonstrates how MongoDB, HBase, and Apache Spark support complementary analytical workloads.

MongoDB stores business documents including product catalog records, user profiles, and transactions with embedded line items. HBase stores time-oriented browsing sessions for efficient retrieval of one user's activity. Spark cleans the raw JSON, creates typed Parquet datasets, runs SQL analytics, and joins the logical HBase and MongoDB sources for an integrated funnel analysis.

Component	Primary responsibility
MongoDB	Nested catalog, user, and transaction documents; document aggregations.
HBase	Time-series browsing sessions retrieved by user/time prefix.
Spark	Cleaning, transformation, SQL analytics, joins, and static visualizations.

Table 1: Responsibility of each platform component.

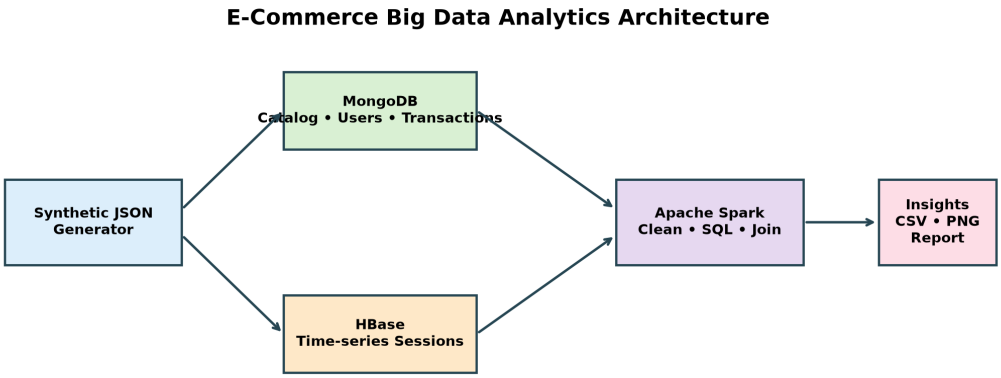


Figure 1: System architecture and flow of data from generation to reporting.

2 Data modelling and storage decisions

2.1 MongoDB

MongoDB database `ecommerce_analytics` contains category documents with embedded subcategories, product documents with price history, user documents with a summarized purchase history, and transaction documents with embedded item lines. Embedding is appropriate because the related fields are usually created and read together. Indexes were created for category, user/time, transaction time, and item product/category analytical filters.

Collection	Key modelling decision
`categories`	Small category hierarchy embedded in each category document.
`products`	Product details, stock, category snapshot, and bounded price history.
`users`	Profile plus embedded order/spend/preferred-category summary.
`transactions`	Header and historical item lines embedded in one purchase document.

Table 2: MongoDB collections and their key modelling decisions.

All 25 categories, 5,000 products, 10,000 users, and 50,000 transactions were loaded. The two implemented aggregation pipelines identify top products by recognized revenue and segment customers by purchase behavior. The load and index operation took 3.413 seconds.

2.2 HBase

The `user_sessions` HBase row key is `user_id|ISO-8601-start_time|session_id`. It co-locates one user's sessions and preserves chronological order for an efficient prefix scan. Column families `meta`, `device`, `geo`, and `activity` group related attributes. The `product_daily_metrics` table uses `product_id|YYYY-MM-DD` for future daily performance facts.

A deterministic 10,000-session representative subset was loaded and verified. The example prefix scan for `user_005955` returned the user's stored session data, including device, referrer, browsing counts, and conversion status.

3 Spark batch processing and SQL analytics

The Spark batch application standardizes IDs, parses timestamps, casts numerical fields, filters invalid records, explodes nested transaction items, and joins items to known products. The resulting Parquet datasets contain 10,000 cleaned users, 5,000 products, and 109,863 transaction-item rows; 92,216 transaction items have recognized statuses (completed, shipped, or delivered).

A co-purchase recommendation indicator self-joins recognized transaction items on `transaction_id`. The condition product A $<$ product B prevents mirrored pairs. Spark produced 68,565 unique pairs. Since the generated catalog is large and baskets are mostly random, the top observed count of two represents a weak demonstration signal rather than a production-quality recommendation.

Spark SQL registers logical MongoDB user/product/transaction views and HBase session/page-view views. Query one joins engagement measures to purchase value by state and value segment. Query two calculates category product-detail views, recognized purchases, units, revenue, and purchases per 100 views. The sales numerator is restricted to the HBase session cohort, maintaining a valid shared denominator.

4 Integrated analytics workflow

Business question: Which device type and acquisition channel create the strongest progression from browsing, through cart activity, to a recognized purchase?

Spark reads the reproducible JSON sample loaded into HBase and cleaned MongoDB-shaped transaction/product Parquet records. It derives browsing and cart stages from HBase conversion status, filters MongoDB records to recognized statuses, and joins both sources by `session_id`. A left join preserves non-purchasing browsing sessions. The final aggregation returns funnel counts, conversion rates, orders, units, revenue, and revenue per purchasing session by device type and referrer.

The integrated result contains 10,000 browsing sessions, 6,517 cart sessions, and 2,031 sessions matched to recognized purchases. HBase is appropriate for session-event evidence, MongoDB for flexible purchase documents, and Spark for the cross-system join and aggregate computation.

5 Visualizations and business insights

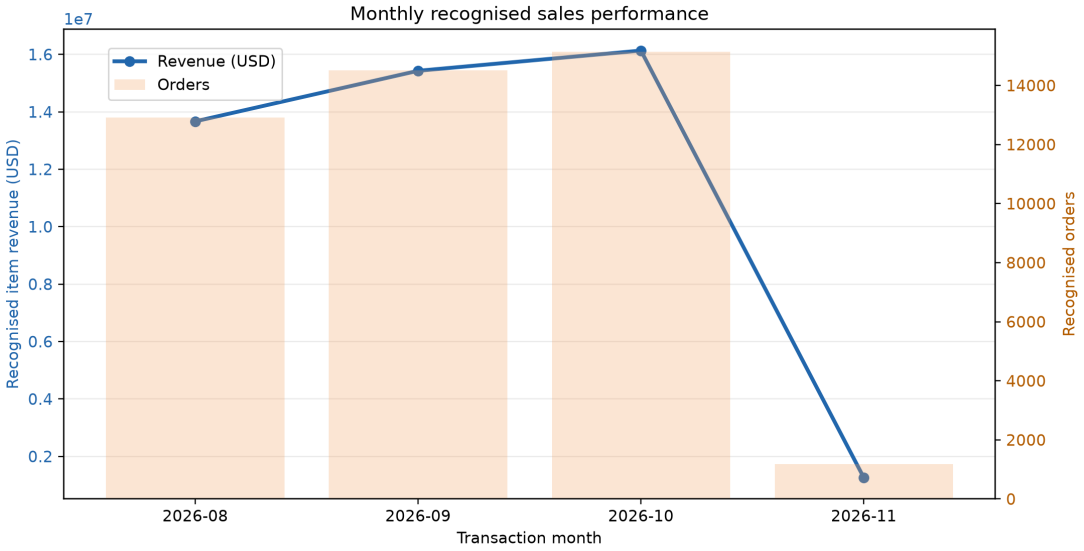


Figure 2: Monthly recognized item revenue and recognized order volume.

Revenue and order volume should be interpreted together. A change in revenue without the same change in order volume may indicate a change in average basket value rather than customer volume.

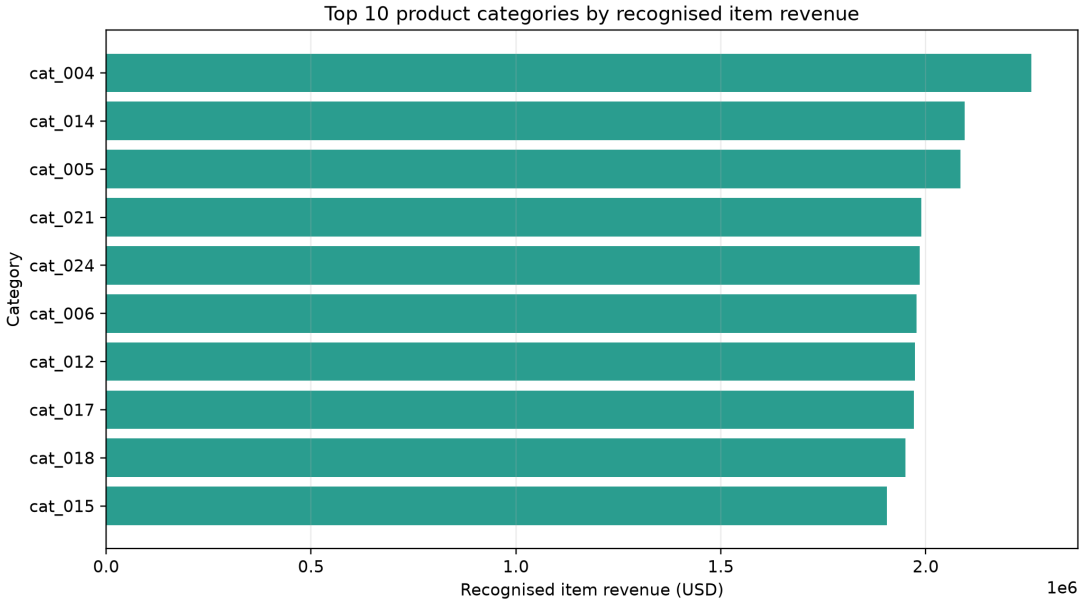


Figure 3: Top ten product categories by recognized item revenue.

High-revenue categories are candidates for inventory monitoring and marketing attention. The measure is revenue, not profit, because product cost, returns, delivery cost, and all discount effects are outside this simplified model.

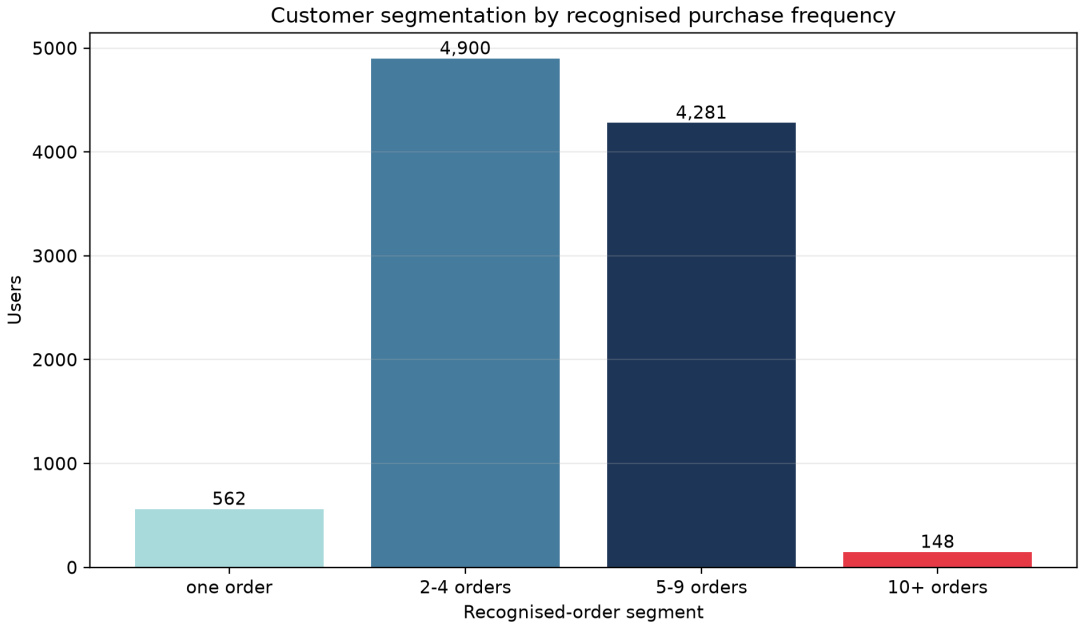


Figure 4: Customer segmentation by recognized purchase frequency.

The frequency distribution distinguishes one-time buyers from repeat customers. This can support retention and re-engagement analysis, although the synthetic data does not establish the causes of repeat purchasing.

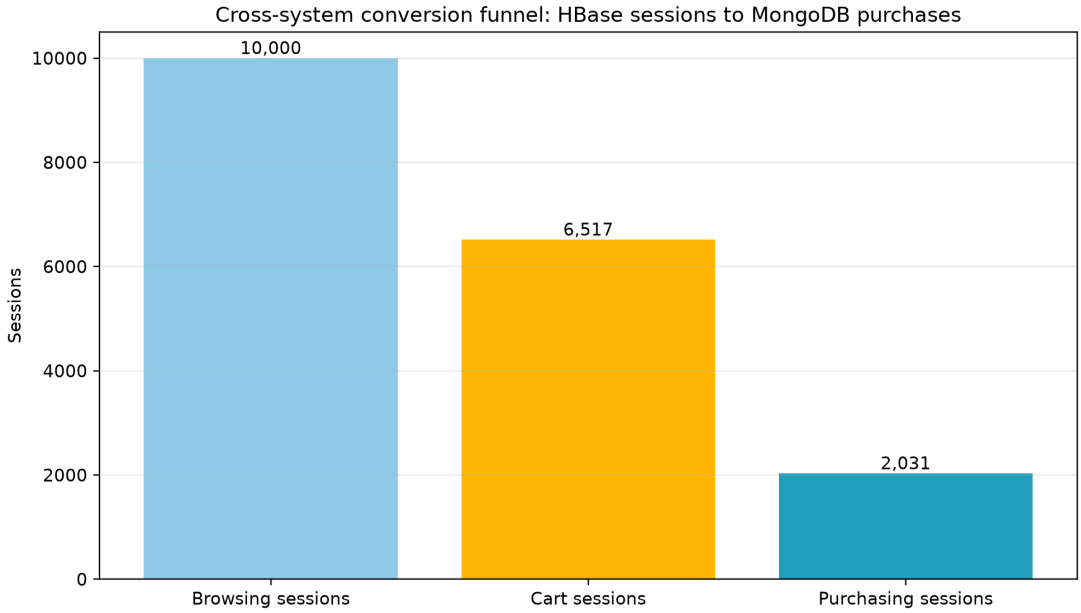


Figure 5: Cross-system funnel from HBase browsing evidence to MongoDB recognized purchases.

The funnel shows substantial drop-off after cart activity. It identifies the checkout stage as a worthwhile investigation point, but it does not prove why users abandoned a cart.

6 Scalability, limitations, and future work

The laptop implementation uses Spark `local[2]` with eight shuffle partitions. Typed Parquet avoids repeated JSON parsing; early recognized-status filtering reduces join input; and deduplication plus product A $<$ product B reduces recommendation self-join output. Production deployment would scale executors and partitions, partition transaction Parquet by date, use time-bounded HBase scans, and possibly add native Spark connectors.

The records are synthetic and largely random, so results show method demonstrations and associations rather

than causal business conclusions. Only a 10,000-session HBase subset was loaded. Future work could load the full stream, incorporate refunds and product costs, create daily HBase product metrics, and use richer behavioral features in a recommendation model.

7 Reproducibility

The repository README provides the commands required to regenerate the raw data, start MongoDB/HBase, load both systems, run Spark processing, run the integrated funnel, and create all visualizations. Key outputs and full query details are retained in JSON summary files. Screenshots of MongoDB, HBase, Spark SQL, integration, and visualization completion should be included as appendices or selected evidence if required by the course instructions.

8 Conclusion

This project demonstrates that an e-commerce analytics platform benefits from using different big-data technologies for the tasks they handle best. MongoDB provided a flexible document model for catalog, user, and transaction data; HBase provided an efficient key-oriented design for time-series browsing sessions; and Apache Spark provided scalable-style batch processing, SQL analysis, and cross-system integration.

The completed analysis produced cleaned Parquet facts, MongoDB aggregation results, HBase session retrieval, recommendation indicators, category and customer analytics, and an integrated funnel from browsing to recognized purchase. In particular, the shared 10,000-session cohort made it possible to calculate a consistent cross-system funnel of 10,000 browsing sessions, 6,517 cart sessions, and 2,031 recognized-purchase sessions.

Although the data is synthetic and the local execution is not a production cluster, the solution establishes a reproducible architecture and a clear path to scale: distributed storage, larger HBase scans, Spark connectors, incremental aggregates, and richer behavioural models. The results illustrate how combining document, wide-column, and distributed-processing technologies can turn operational e-commerce records into actionable analytical evidence.